Set Topology

Lecture 08: Continuous Maps

Zamir Hussain
Department of Mathematics
University of Wah

April 7, 2026

What is a Continuous Map?

In calculus, a function is continuous if you can draw its graph without lifting your pen. In topology, we generalize this idea:

A continuous map is a function that "preserves nearness" between two topological spaces.

The key tool is the **pre-image**: given $f : X \to Y$ and a set $V \subseteq Y$,

$$f^{-1}(V) = \{x \in X \mid f(x) \in V\}.$$

1 The Definition of Continuity

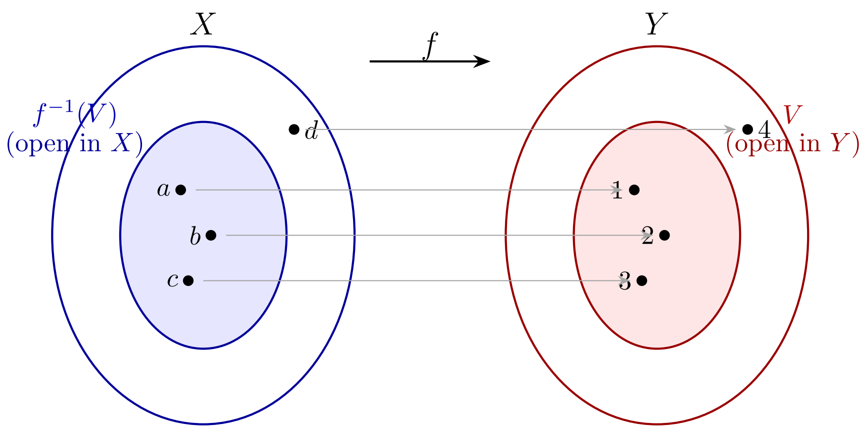
Definition: Continuous Map

Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. A function $f : X \to Y$ is **continuous** if:

$$\text{For every open set } V \in \mathcal{T}_Y, \quad f^{-1}(V) \in \mathcal{T}_X.$$

In one line: *Pre-images of open sets are open.*

1.1 Visualizing the Pre-image



Key identity: For any $A \subseteq Y$, $\quad f^{-1}(Y \setminus A) = X \setminus f^{-1}(A)$.

2 Examples: Continuous and Non-Continuous

Example: Continuous function on finite sets

Let $X = \{a, b, c\}$, $\mathcal{T}_X = \{\emptyset, \{a\}, \{b\}, \{a, b\}, X\}$.
Let $Y = \{1, 2, 3\}$, $\mathcal{T}_Y = \{\emptyset, \{1, 2\}, Y\}$.
Define f by $f(a) = 1$, $f(b) = 2$, $f(c) = 3$.

Step: Check the pre-image of every open set in Y.

Open set $V \in \mathcal{T}_Y$	Pre-image $f^{-1}(V)$	Open in X?
$\emptyset$	$\emptyset$	✓
$\{1, 2\}$	$\{a, b\}$	✓ $(\{a, b\} \in \mathcal{T}_X)$
Y	X	✓

Conclusion: Every pre-image is open $\Rightarrow$ f is **continuous**.

Example: Non-continuous function on finite sets

Let $X = \{a, b, c\}$, $\mathcal{T}_X = \{\emptyset, \{a\}, \{c\}, \{a, c\}, X\}$.
Let $Y = \{1, 2, 3\}$, $\mathcal{T}_Y = \{\emptyset, \{1, 2\}, Y\}$.
Define h by $h(a) = 1$, $h(b) = 2$, $h(c) = 3$.

Step: Check the open set $V = \{1, 2\}$.

$$h^{-1}(\{1, 2\}) = \{a, b\}.$$

But $\{a, b\} \notin \mathcal{T}_X$. One pre-image is not open.
Conclusion: h is **not continuous**.

Example: Step function on $\mathbb{R}$

Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0. \end{cases}$$

Step: Pick the open set $V = \left(\frac{1}{2}, \frac{3}{2}\right)$.

$$f^{-1}(V) = \{x \in \mathbb{R} \mid f(x) > \tfrac{1}{2}\} = [0, \infty).$$

The set $[0, \infty)$ is *not open* in $\mathbb{R}$ (no open interval around 0 stays inside $[0, \infty)$).
Conclusion: f is **not continuous** at $x = 0$.

3 Three Equivalent Formulations

The three rows below describe the same condition in three different languages.

Formulation	Statement
Open Sets (definition)	f is continuous iff $f^{-1}(V)$ is open in X for every open $V \subseteq Y$.
Closed Sets	f is continuous iff $f^{-1}(C)$ is closed in X for every closed $C \subseteq Y$.
Sequential (metric spaces)	f is continuous at x iff $x_n \to x$ implies $f(x_n) \to f(x)$.

Key Remarks

Three things to keep distinct:

1. **Continuity $\neq$ openness of f.** Continuity says *pre-images* of open sets are open. A map is *open* if *images* of open sets are open. These are different.
2. **Continuity $\neq$ continuity of f^{-1}.** A continuous bijection need not have a continuous inverse. A bijection where both f and f^{-1} are continuous is called a **homeomorphism**.
3. **Pre-image $\neq$ image.** $f^{-1}(V) = \{x \mid f(x) \in V\}$. $f(A) = \{f(x) \mid x \in A\}$. Do not mix them in proofs.

4 Connection to the ε-δ Definition

In calculus, continuity at x_0 means: for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

This is a special case of the topological definition where open sets are open balls $B_\varepsilon(x_0)$.

Example: $f(x)$

ε-δ approach. Fix $x_0 \in \mathbb{R}$, $\varepsilon > 0$. Choose $\delta = \varepsilon/2$. Then:

$$|x - x_0| < \delta \implies |2x - 2x_0| = 2|x - x_0| < 2 \cdot \tfrac{\varepsilon}{2} = \varepsilon. \quad \checkmark$$

Topological approach. Let $V \subseteq \mathbb{R}$ be open, and take $x_0 \in f^{-1}(V)$, so $2x_0 \in V$.

Step 1. Since V is open, $\exists\, \varepsilon > 0$ such that $(2x_0 - \varepsilon,\, 2x_0 + \varepsilon) \subseteq V$.

Step 2. Set $\delta = \varepsilon/2$.

Step 3. For $x \in (x_0 - \delta,\, x_0 + \delta)$: $\quad 2x \in (2x_0 - \varepsilon,\, 2x_0 + \varepsilon) \subseteq V, \quad$ so $x \in f^{-1}(V)$.

Step 4. Hence $(x_0 - \delta, x_0 + \delta) \subseteq f^{-1}(V)$, so $f^{-1}(V)$ is open. $\checkmark$

Connection: The $\delta = \varepsilon/2$ in both approaches is identical. The topological definition just drops the numbers and keeps only the structural fact that an open neighborhood exists.

Proposition/Theorem: Equivalence with ε-δ

For functions between metric spaces, the topological and ε-δ definitions of continuity are equivalent.

Proof sketch.
(ε-$\delta \Rightarrow$ topological): Let $V \subseteq Y$ be open, $x_0 \in f^{-1}(V)$. Then $\exists\, \varepsilon > 0$ with $B_\varepsilon(f(x_0)) \subseteq V$. By ε-δ, $\exists\, \delta > 0$ with $f(B_\delta(x_0)) \subseteq B_\varepsilon(f(x_0)) \subseteq V$. So $B_\delta(x_0) \subseteq f^{-1}(V)$, making $f^{-1}(V)$ open.
(topological $\Rightarrow \varepsilon$-δ): Fix x_0, $\varepsilon > 0$. Then $B_\varepsilon(f(x_0))$ is open, so $f^{-1}(B_\varepsilon(f(x_0)))$ is open in X. It contains x_0, so $\exists\, \delta > 0$ with $B_\delta(x_0) \subseteq f^{-1}(B_\varepsilon(f(x_0)))$. ■

5 Equivalent Definition via Closed Sets

Proposition/Theorem: Continuity via Closed Sets

$f : X \to Y$ is continuous $\iff$ for every closed set $C \subseteq Y$, the pre-image $f^{-1}(C)$ is closed in X.

Proof. Use the identity $f^{-1}(Y \setminus A) = X \setminus f^{-1}(A)$.
($\Rightarrow$): Let C be closed in Y. Then $Y \setminus C$ is open. By continuity, $f^{-1}(Y \setminus C) = X \setminus f^{-1}(C)$ is open. So $f^{-1}(C)$ is closed.
($\Leftarrow$): Let V be open in Y. Then $Y \setminus V$ is closed, so $X \setminus f^{-1}(V)$ is closed. Hence $f^{-1}(V)$ is open. ■

Example: Closed-set check for $f(x)$

- **Open-set check:** $f^{-1}((1, 4)) = (-2, -1) \cup (1, 2)$ — open. ✓
- **Closed-set check:** $f^{-1}([1, 4]) = [-2, -1] \cup [1, 2]$ — closed. ✓

Both checks confirm $f(x) = x^2$ is continuous.

6 Properties of Continuous Maps

Proposition/Theorem: Four Stability Properties

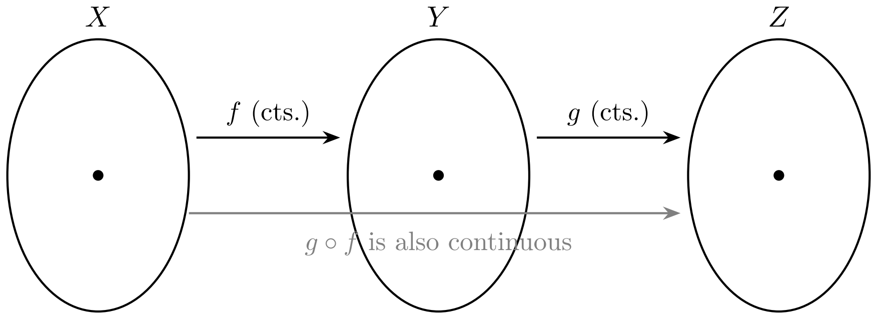
Let $f : X \to Y$ and $g : Y \to Z$.

(a) **Constant maps:** $f(x) = y_0$ for all $x \Rightarrow f$ is continuous.

(b) **Identity:** $\mathrm{id}_X : X \to X$ is continuous.

(c) **Composition:** f and g continuous $\Rightarrow g \circ f : X \to Z$ is continuous.

(d) **Restriction:** f continuous, $U \subseteq X$ open $\Rightarrow f|_U : U \to Y$ is continuous.



Proofs.

(a) If $y_0 \in V$ then $f^{-1}(V) = X$; if $y_0 \notin V$ then $f^{-1}(V) = \emptyset$. Both open. ✓

(b) $\mathrm{id}_X^{-1}(U) = U$ for every $U \subseteq X$. ✓

(c) $(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V))$. g continuous $\Rightarrow$ $g^{-1}(V)$ open in Y. f continuous $\Rightarrow$ $f^{-1}(g^{-1}(V))$ open in X. ✓

(d) $(f|_U)^{-1}(V) = U \cap f^{-1}(V)$. Both U and $f^{-1}(V)$ are open in X, so their intersection is open. ✓

7 Sequential Continuity

Proposition/Theorem: Continuous maps preserve sequence limits

If $f : X \to Y$ is continuous and $x_n \to x$ in X, then $f(x_n) \to f(x)$ in Y.
Proof.

Step 1. Let V be any open neighborhood of $f(x)$ in Y.

Step 2. f is continuous, so $f^{-1}(V)$ is open in X and contains x.

Step 3. $x_n \to x$, so $\exists\, N$ such that $x_n \in f^{-1}(V)$ for all $n > N$.

Step 4. $x_n \in f^{-1}(V)$ means $f(x_n) \in V$.

Step 5. So $f(x_n) \in V$ for all $n > N$. Hence $f(x_n) \to f(x)$. ■

Example: Sequential check for $f(x)$

- $x_n = 1/n \to 0$: $f(x_n) = 1/n^2 \to 0 = f(0)$. ✓
- $x_n = 1 + 1/n \to 1$: $f(x_n) = (1 + 1/n)^2 \to 1 = f(1)$. ✓

Note: In metric spaces (first-countable spaces), the converse also holds: sequential continuity implies continuity. In general topological spaces it does not.

8 Practice Problems

1. Let $X = \{1, 2, 3, 4\}$ with the cofinite topology, $Y = \{a, b\}$ with $\mathcal{T}_Y = \{\emptyset, \{a\}, Y\}$. Define $f(1) = f(2) = a$, $f(3) = f(4) = b$. Is f continuous? Justify by checking every open set in Y.

2. Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$.

(a) Show $f^{-1}((1,4))$ is open in $\mathbb{R}$.
(b) Show $f^{-1}([1,4])$ is closed in $\mathbb{R}$.
(c) Using sequential continuity, verify $f(x_n) \to f(2)$ when $x_n = 2 + 1/n$.

3. **(Images of closed sets)** Let $f(x) = e^{-x}$ on $\mathbb{R}$, $A = [0, \infty)$.

    (a) Compute $f(A)$ explicitly.
    (b) Is $f(A)$ closed in $\mathbb{R}$? (Is $0 \in f(A)$? Is 0 a limit point of $f(A)$?)
    (c) Conclude: does continuity imply images of closed sets are closed?

4. Let X carry the discrete topology. Prove every $f : X \to Y$ is continuous.

5. Let Y carry the indiscrete topology. Prove every $f : X \to Y$ is continuous.

6. **(Challenge)** Let $f : \mathbb{R} \to \mathbb{R}$ be the step function (Section 2, Example 3). Show f is continuous at every $x_0 \neq 0$ by finding an explicit open set $U \ni x_0$ on which f is constant.

Summary

- $f : X \to Y$ is **continuous** $\iff$ pre-image of every open set is open.
- **Equivalent:** pre-image of every closed set is closed.
- **Equivalent (metric spaces):** $x_n \to x \Rightarrow f(x_n) \to f(x)$.
- Constant maps, identity, compositions, and restrictions to open sets are all continuous.
- **Do not confuse:** continuity $\neq$ openness of f; and continuity of $f \neq$ continuity of f^{-1}.
- The ε-δ definition is a special case: replace open balls by open sets.

Proposition/Theorem: Further Reading

Munkres, *Topology* (2nd ed.): Section 18 “Continuous Functions”.